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A genetic algorithm optimized hybrid model for agricultural price forecasting based on VMD and LSTM network

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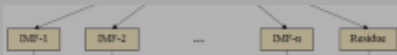
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oil, and soybean oil. Performance is assessed through root mean square error (RMSE), mean absolute percentage error (MAPE), and directional prediction statistics ($\{D\}_{stat}$). VMD-LSTM reduces RMSE by 56.93%, 21.83%, and 27.00% and MAPE by 44%, 21.67%, and 25.85% for maize, palm oil, and soybean oil, respectively, compared to the next best CEEMDAN-LSTM. TOPSIS and Diebold-Mariano test also confirm the better prediction accuracy of VMD-LSTM. The proposed model can be a better tool for agricultural price forecasting, supporting decision-making for farmers, traders, and policymakers.

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which can significantly affect crop yields; seasonal patterns in production, where certain crops are grown and harvested during specific times of the year; and derived demand, which refers to the demand for agricultural products driven by their use in other industries. Additionally, imperfections in the market, such as lack of transparency or unequal access to resources, play a role. Economic globalization further adds complexity by exposing the market to international trade dynamics and competition. Lastly, administrative regulations, such as government policies and trade restrictions, have a direct impact on market operations and pricing. These factors make the price series extremely complex, nonlinear, nonstationary, and chaotic, making price prediction an extremely challenging task². Accurate and reliable forecasts of agricultural commodity prices are critical for balancing the interests of producers and consumers, promoting market stability, and enabling informed decision-making. In this regard, the research community has put enormous efforts in order to improve the accuracy of price

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modelling, classification and prediction by own. DL approaches improve traditional ANNs by enhancing their complexity and precision, leading to successful applications in various fields⁹. An increasingly promising deep learning technique for price forecasting is the long short-term memory (LSTM) model, which excels at learning temporal sequences and is well-suited. LSTM belongs to the family of recurrent neural networks and employs sophisticated gated mechanisms to retain information over extended periods, effectively addressing the vanishing gradient problem¹⁰. It also handles long-term dependencies, which are crucial in price forecasting. Overall, the potential of DL methods, particularly LSTM, in time series forecasting offers exciting prospects for improving accuracy and precision in agricultural commodity markets. Jaiswal et al. (2021)¹¹ demonstrate superior forecasting performance of the LSTM model over ANN and ARIMA models for international palm oil and maize prices.

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ARIMA, ANN, and support vector machine (SVM) models to predict the decomposed components. Zhang et al. (2014)²³ proposed an EEMD-based hybrid model that integrated generalized autoregressive conditional heteroscedasticity (GARCH) and LSSVM to forecast crude oil prices. Prasad et al. (2018)²⁴ showcased the superiority of a hybrid model that combines CEEMDAN and an extreme learning machine (ELM) for soil moisture forecasting. Cao et al. (2019)²⁵ combined CEEMDAN with LSTM for stock market prices forecasting. Meanwhile EMD and its variants have been applied successfully in some time series forecasting applications, but still have many limitations^{26,27,28}. One major challenge of these methods is the issue of boundary effects, which occur when the decomposition process leads to the creation of artificial extrema at the boundaries of the data, thus introducing errors into the final prediction results. In addition, EMD and its variants are prone to mode mixing, where the resulting decomposed components may contain mixed information from multiple underlying

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hybrid models. In recent years, various studies have been conducted to develop hybrid models based on combining VMD and LSTM^{31,35,36}. Despite the advantages of VMD and LSTM, the accuracy of these hybrid models remains unsatisfactory due to the lack of standard rules for selecting the hyperparameters of both tools. Therefore, optimization techniques such as genetic algorithm (GA) have gained attention in recent studies to determine hyperparameters' optimal values³⁷. The GA is a powerful optimization technique that can search the vast solution space of hyperparameters and find the best combination of hyperparameters to minimize the prediction error of the hybrid model. Using GA, the hybrid models can capture the sharp fluctuations in the agricultural price series and provide more accurate and reliable forecasts for agricultural stakeholders.

This paper proposes a novel GA-optimized hybrid model, VMD-LSTM, that harnesses the unique advantages of GA, VMD, and LSTM techniques, each contributing to

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and the prediction strength of LSTM. Here, GA-optimized VMD disintegrates a highly complex agricultural price series into a set of IMFs with minimum decomposition loss compared to existing EMD variants. Further, for modelling each IMF, we utilized GA-optimized LSTM, aiming to enhance the overall prediction accuracy of our hybrid model.

2) We use GA as an optimization technique to solve the parameter selection problem for VMD and LSTM.

3) We use three real agricultural price series to evaluate our proposed models empirically. The results demonstrate the superior performance of our proposed hybrid model compared to baseline models concerning both level and direction prediction accuracy. To comprehensively understand and rank all the models, we

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bandwidth of the central frequency for each intrinsic mode function (IMF). VMD decomposition yields modes with a unique property of sparsity, where each mode is mainly centred around a specific pulsation in the frequency spectrum. This characteristic distinguishes VMD from other decomposition techniques. Additionally, VMD offers several advantages, making it superior in handling complex data, particularly nonstationary and nonlinear data. Primarily, the modes are robust to noise, essential when dealing with complex agricultural price data consisting of multi-frequency series.

Additionally, VMD has faster convergence and better accuracy. These characteristics of VMD make it highly suitable for addressing complex agricultural price data consisting of multi-frequency signals. The following steps are employed to estimate the bandwidth of a mode:

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where $y(t)$ is the original price series, $\{c_j\}_{j=1}^n$ is the set of modes, $\{w_j\}_{j=1}^n$ is the of central frequencies, $\delta(t)$ is the impulse function, ∂_t is the partial derivative of time t , n is the number of modes, $*$ denotes convolution operation, $\parallel$ denotes norm processing, and $i=\sqrt{-1}$. Lagrangian multipliers $\lambda(t)$ and quadratic penalty terms are used to transform a constraint problem into an unconstrained problem that is easy to solve:

$$L(\{c_j\}, \{\omega_j\}, \lambda) = \alpha \sum_j \parallel \partial_t \left[\delta(t) + \frac{i}{\pi t} \right] c_j(t) \right\| e^{-i\omega_j t} \parallel \frac{0}{2} \parallel^2 + \parallel y(t) - \sum_j c_j(t) \parallel \frac{0}{2}$$

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- 2) Evaluation: Each set of VMD parameters decomposes the time series into its constituent modes (IMFs). After taking the initial parameter, VMD's decomposed IMFs follow as: $\left(\stackrel{\prime}{y}\right)\left(t\right)=\sum_{j=1}^n\left\{c_j\right\}\left(t\right)$. Ideally, the sum of all IMFs should be the same as the original series as: $\left(\stackrel{\prime}{y}\right)\left(t\right)=y\left(t\right)$. However, it does not generally hold, as the VMD causes decomposition loss. Hence, we define the fitness function which minimizes decomposition loss, i.e. the ratio of residual energy $\left\{r_{res}\right\}=\frac{\sum_{t=1}^T\left|\frac{y\left(t\right)-\sum_{j=1}^n\left\{c_j\right\}\left(t\right)}{y\left(t\right)}\right|^2}{T}$, where T is the total observations.
- 3) Selection: The algorithm selects the best-fit sets of VMD parameters from the current population to be parents for the next generation. The selection process is based on the fitness function values of the solutions. Table 1 shows the

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Replacement: In this step, the newly generated offspring solutions are introduced into the population by replacing the least-fit solutions. This process ensures that the weaker individuals, which are less likely to contribute to an optimal solution, are removed. As a result, a new generation of VMD parameters is created, gradually steering the population towards better solutions.

7) Termination: the algorithm stops when a termination criterion is met.

Once all hyperparameters are optimized, the obtained IMFs are modelled individually through the LSTM model to help improve the prediction accuracy of an agriculture price series.

Table 1 Range of hyperparameters for genetic Algorithm-Optimized variational

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$$f_t = \sigma(W_f \bullet [h_{t-1}, y_t] + b_f)$$

The subsequent step involves storing new or relevant information in the cell state. This is accomplished through another gate that combines a sigmoid function known as the “input gate” and a tanh layer, as expressed in the following equations:

$$i_t = \sigma(W_i \bullet [h_{t-1}, y_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c \bullet [h_{t-1}, y_t] + b_c)$$

Lastly, the output is determined based on the updated cell state $\tilde{C}_t$, utilizing a sigmoid layer and a *tanh* layer, which selectively output the relevant parts as expressed in the following equations:

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leading to better prediction performance of individual IMF. GA-based optimization can help overcome the limitations of manual tuning of significant hyperparameters⁴². It can also help to avoid overfitting by selecting the optimal combination of hyperparameters.

The genetic algorithm operates by generating an initial set of possible solutions. This population of solutions is then iteratively improved through a process that mimics natural selection, where the fittest solutions are selected and recombined to create new, potentially better solutions in subsequent generations. Each individual in the population represents a set of hyperparameters for the LSTM model. The range of these hyperparameters is shown in Table 2. Overall, a genetic algorithm-based LSTM model can be a powerful tool for optimizing the performance of LSTM models on individual IMF prediction.

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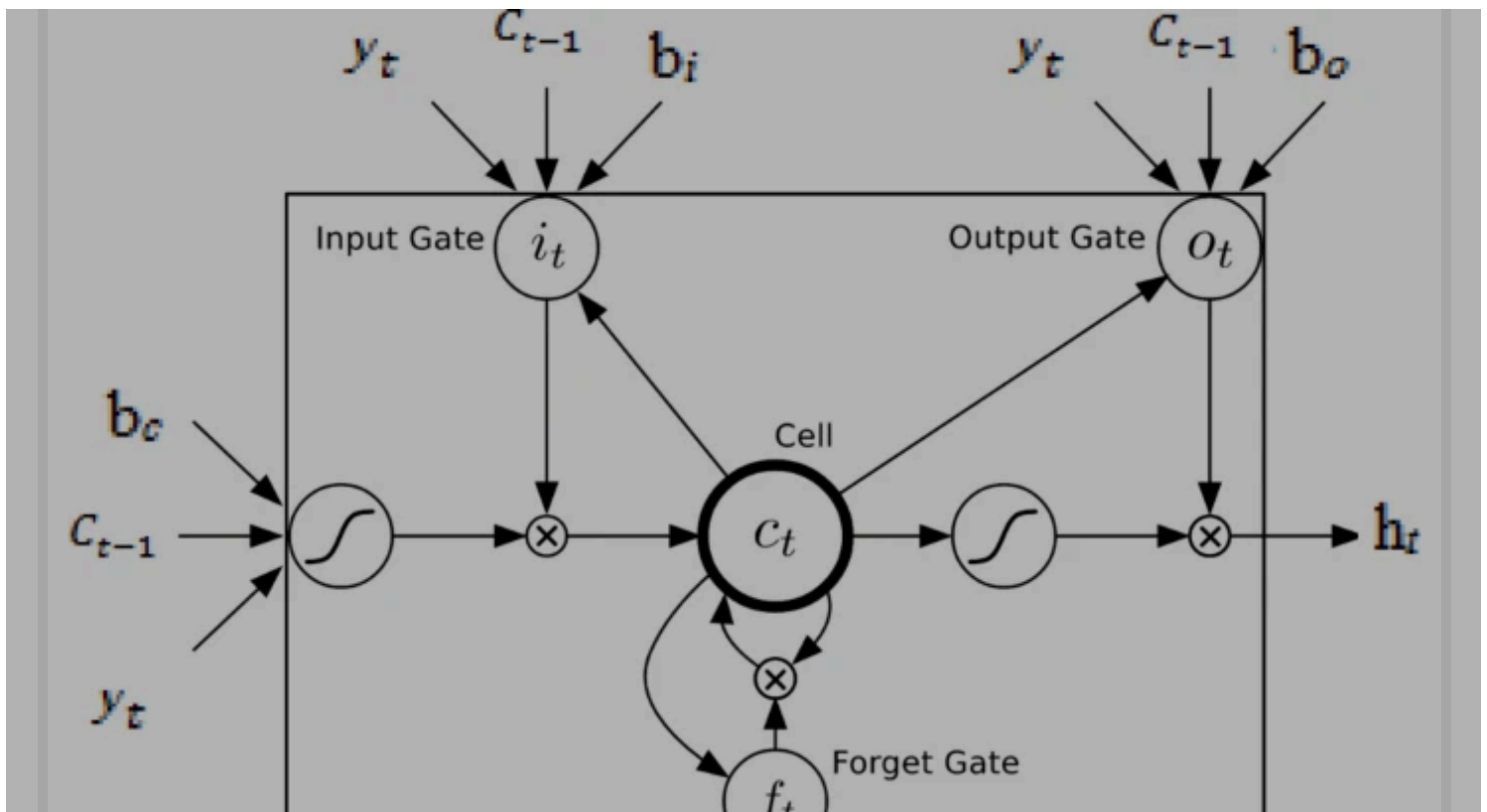
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structure with stable fluctuations. This step makes it easier to identify and model the underlying trends and behaviors in the price data.

2. **Individual prediction:** Each decomposed IMF is divided into training and testing datasets, with a portion of the training data reserved for validation. This split ensures the forecasting model's ability to generalize to unseen data. The GA-optimized LSTM model is employed to predict each mode individually. This model is particularly effective in capturing the nonlinear relationships and long-term dependencies within the IMFs, providing accurate predictions for each frequency component.

3. **Ensemble prediction:** The final forecast of the original price series is obtained by summing up the predicted values of all Intrinsic Mode Functions (IMFs) together

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Flowchart of the proposed genetic algorithm optimized hybrid model for agricultural price forecasting.

VMD is instrumental in decomposing complex, nonstationary price series into IMFs, effectively mitigating noise and extracting essential price dynamics. Unlike existing VMD-based models that rely on manually tuned parameters, our model employs GA to optimize the selection of VMD parameters [number of modes (n), balancing parameter of the data-fidelity constrain (α), tolerance of convergence criterion (τ), and time-step of the dual ascent (ϵ)]. This automated optimization process enhances the decomposition's precision and adaptability to various datasets, leading to more reliable and granular representations of price series. Each decomposed IMF is individually modeled using GA-optimized LSTM networks, ensuring that each IMF's temporal

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TOPSIS method⁴³ is employed, which ranks each model by giving weights after normalizing the decision matrix of all evaluation criteria and calculating the geometric distance between the different models. The following are the forecasting evaluation criteria for comparing the proposed model with other models:-.

$$RMSE = \sqrt{\frac{1}{h} \sum_{t=1}^h (\left| y(t) - \widehat{y}(t) \right|)^2}$$

(1)

$$MAPE = \frac{1}{h} \sum_{t=1}^h \left| \frac{y(t) - \widehat{y}(t)}{y(t)} \right|$$

(2)

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where h is the test size, $\left\{e_{tet}\right\}_{t=1}^h$ and $\left\{e_{ref}\right\}_{t=1}^h$ are errors for the test model and reference model, respectively, g is the loss function, $d=\frac{1}{h}\sum_{t=1}^h\left[g\left(e_{tet}\right)-g\left(e_{ref}\right)\right]$ is the sample mean, $\widehat{V}_{\stackrel{\sim}{d}}=\frac{1}{h}\left[\gamma_0+2\sum_{j=1}^{l-1}\gamma_j\right]$ is the estimate of variance using l -step forecasts and $\gamma_j=\text{cov}\left(d_t,d_{t-j}\right)$ is the estimate of j^{th} autocovariance of $\left[g\left(e_{tet}\right)-g\left(e_{ref}\right)\right]$.

Technique for order of preference by similarity to ideal solution

For a given decision matrix $X=\left(x_{ij}\right)$ and a weight vector $W=\left(w_1, w_2, \dots, w_n\right)$, rank of i^{th} model is defined

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Results and discussion

Data set

This study utilizes data from three monthly international price series: Maize, Palm oil, and Soybean oil (measured in dollars per metric tonne, \$/MT). The data is sourced from the "World Bank Commodity Prices Data" (<https://www.worldbank.org/en/research/commodity-markets>) and spans from January 1960 to December 2021.

The three price series were divided into two sets for analysis. The first set consists of the training and validation sets from January 1960 to December 2020, totalling 732 observations. The second set is the testing set, comprising the remaining 12 observations. The data is visualized using time plots for each price series to begin the time series analysis, as illustrated in Fig. 2. These time plots reveal the nonlinear and

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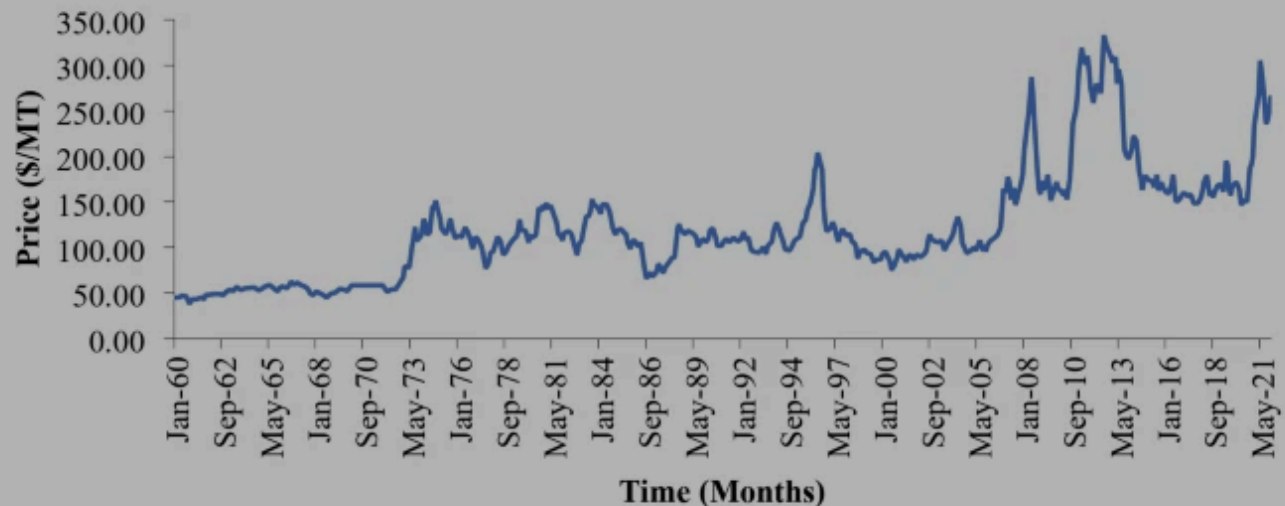
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Maize



Palm oil

1500.00

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Tests for stationarity and linearity

Since agricultural price series are complex and may exhibit nonstationarity and nonlinearity properties, it becomes necessary to test these properties, which can be helpful in the skilful handling of data while fitting the model. The Augmented Dickey-Fuller (ADF)⁴⁴ and Brock-Dechert-Scheinkman (BDS) tests²¹ are used to check the stationarity and linearity characteristics, respectively.

Table 4 shows the ADF test results confirming the nonstationarity of each price series. The test result demonstrates that the price series of maize, palm oil, and soybean oil are

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Decomposition of the agricultural price series

After checking prerequisite conditions for decomposition, i.e. nonlinearity and nonstationarity characteristic of agricultural price series, we use four decomposition tools: EMD, EEMD, CEEMDAN, and VMD. These tools effectively break down the original price series into simpler and more stable components for better analysis. Techniques like Empirical Mode Decomposition (EMD) and its variants—Ensemble Empirical Mode Decomposition (EEMD) and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN)—decompose the series into a set of Intrinsic Mode Functions (IMFs) and a residual. The total number of modes (IMFs and residual) is approximately $\lceil \log_2 T \rceil$, where T is the total data points. Besides the number of modes, the EMD requires two more hyperparameters, i.e., the stopping criterion and the maximum number of siftings. EEMD and CEEMDAN need four additional

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For example, If the number of modes is too small, it may not be sufficient to decompose the original series, resulting in inaccurate predictions adequately. Conversely, if the time series is excessively decomposed, the differences between each component become minimal, leading to reduced accuracy and unnecessary computing overheads. Since the grid search approach to generate the optimum model may be exceedingly time-consuming and tedious, this task can be automated using genetic algorithms that optimize results by selecting candidates from their search space in a random manner and achieving results in less time. We apply binary-coded GA to our model. The hyperparameter values range used in this study are as follows: (i) number of modes: $\lfloor \frac{1}{\sqrt{n}} \rfloor \in \{2, 3, 4, \dots, 25\}$ (ii) balancing parameter of the data-fidelity constrain $\alpha \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ (iii) tolerance of convergence criterion: $\tau \in \{0, 1\}$ and (iv) time-step of the dual ascent: $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$

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decomposed modes through GA-based VMD from high frequency to low frequency. Here, high frequency shows the effect of short-term fluctuations in the market.

In contrast, low frequency represents any particularly significant event affecting the demand-supply equilibrium at that time. For instance, in our case, the two most significant events were observed in 2008 and 2011 which can be observed in Fig. 4 shows spikes around the 580th and 620th observations, respectively. Both events can be attributed to the 2007-08 global food crisis and the rise in biofuel production⁴⁵. The utilization of maize for ethanol fuel escalated from 15% (in 2006) to 40% (in 2012) of the entire U.S. maize production. Furthermore, the Intrinsic Mode Functions (IMFs) obtained from VMD-based decomposition exhibit a more independent and distinct frequency distribution compared to those derived from EMD variants. This enhanced independence can be empirically validated by analyzing the energy evaluation parameter \

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Table 6 Parameters utilized by genetic algorithm for each price series to get optimum results.

Table 7 Variational mode decomposition parameters through genetic algorithm for different price series.

Table 8 Comparison of different decomposition algorithm in terms of energy evaluation parameter ($\|\vec{\theta}\|$)

Fig. 4

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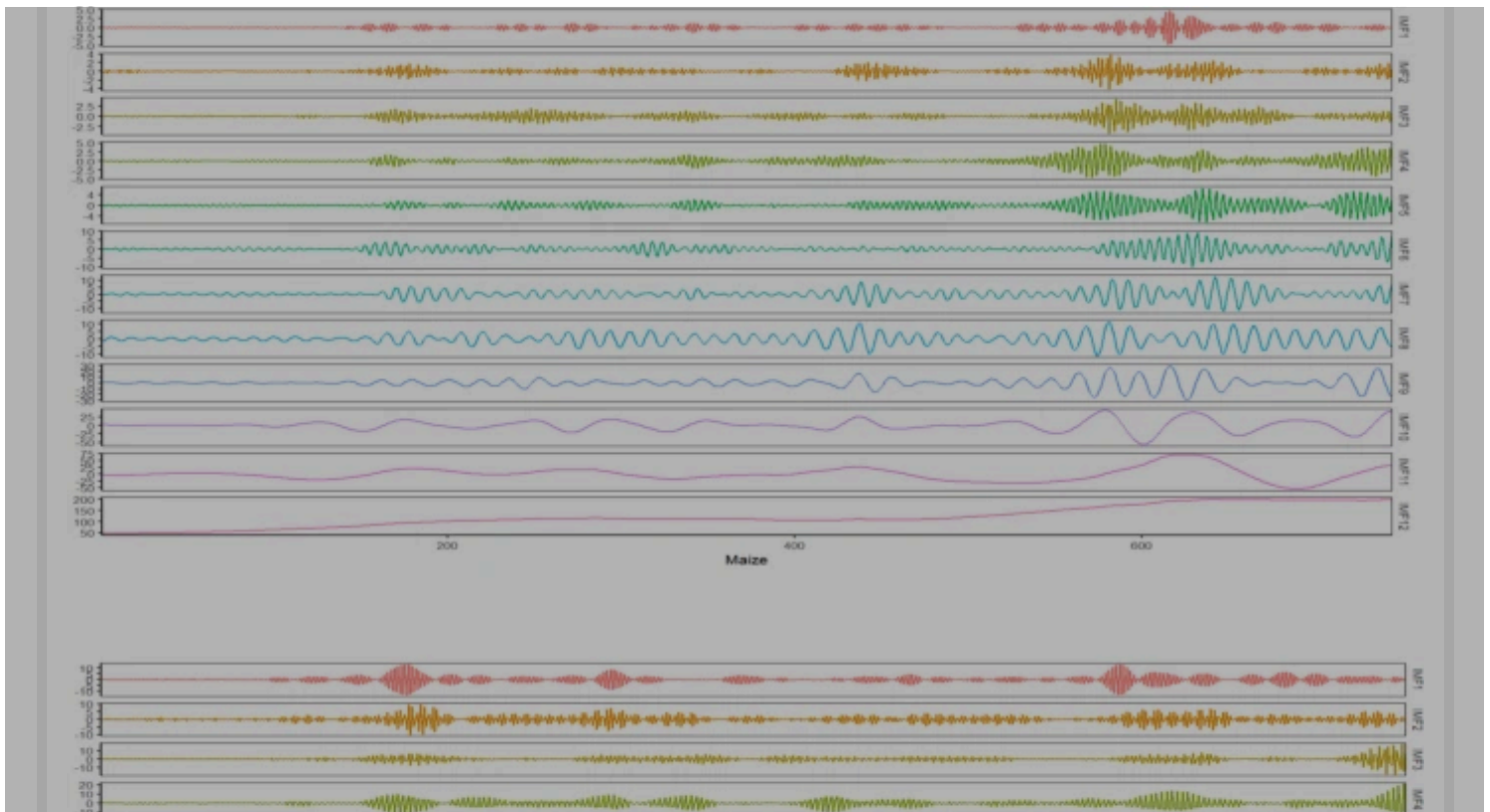
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Forecasting results

The forecasting performance of the GA-optimized VMD-LSTM model is evaluated by comparing it with four other models: LSTM, EMD-LSTM, EEMD-LSTM, and CEEMDAN-LSTM. To facilitate this comparison, we developed two R software packages, named [decompDL³⁸](#) and [TSdeeplearning³⁹](#), which are publicly available on the Comprehensive R Archive Network (CRAN). These packages were specifically designed to support the decomposition and deep learning tasks required for time series forecasting, providing a robust platform for model implementation and evaluation. The comparison is based on international monthly maize, palm, and soybean oil prices.

A standard LSTM model is developed to forecast each price series individually in this comparison. Whereas under the hybrid model (EMD-LSTM, EEMD-LSTM, CEEMDAN-

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model, especially when applied to a variety of datasets. In a similar manner, each Intrinsic Mode Function (IMF) is predicted individually using the LSTM, which effectively captures the specific patterns within each IMF. Ultimately, the forecast for the original price series is obtained by linearly summing the predictions of all the IMFs, providing a comprehensive and accurate forecast for the entire series. Figure 7 demonstrates the forecasting results of each predicted model for the three agriculture price series in the test set. Table 10 quantitatively and Fig. 8 visually display a detailed comparison of the MAPE, RMSE, $\sqrt{\frac{1}{N} \sum_{i=1}^N (D_i - \hat{D}_i)^2}$ and TOPSIS for each prediction model. Table 11 displays the DM test results among individual models (LSTM) and its decomposition-based hybrid models (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM, and VMD-LSTM) for each price series. The following can be concluded from these tables and figures:

Fig. 5

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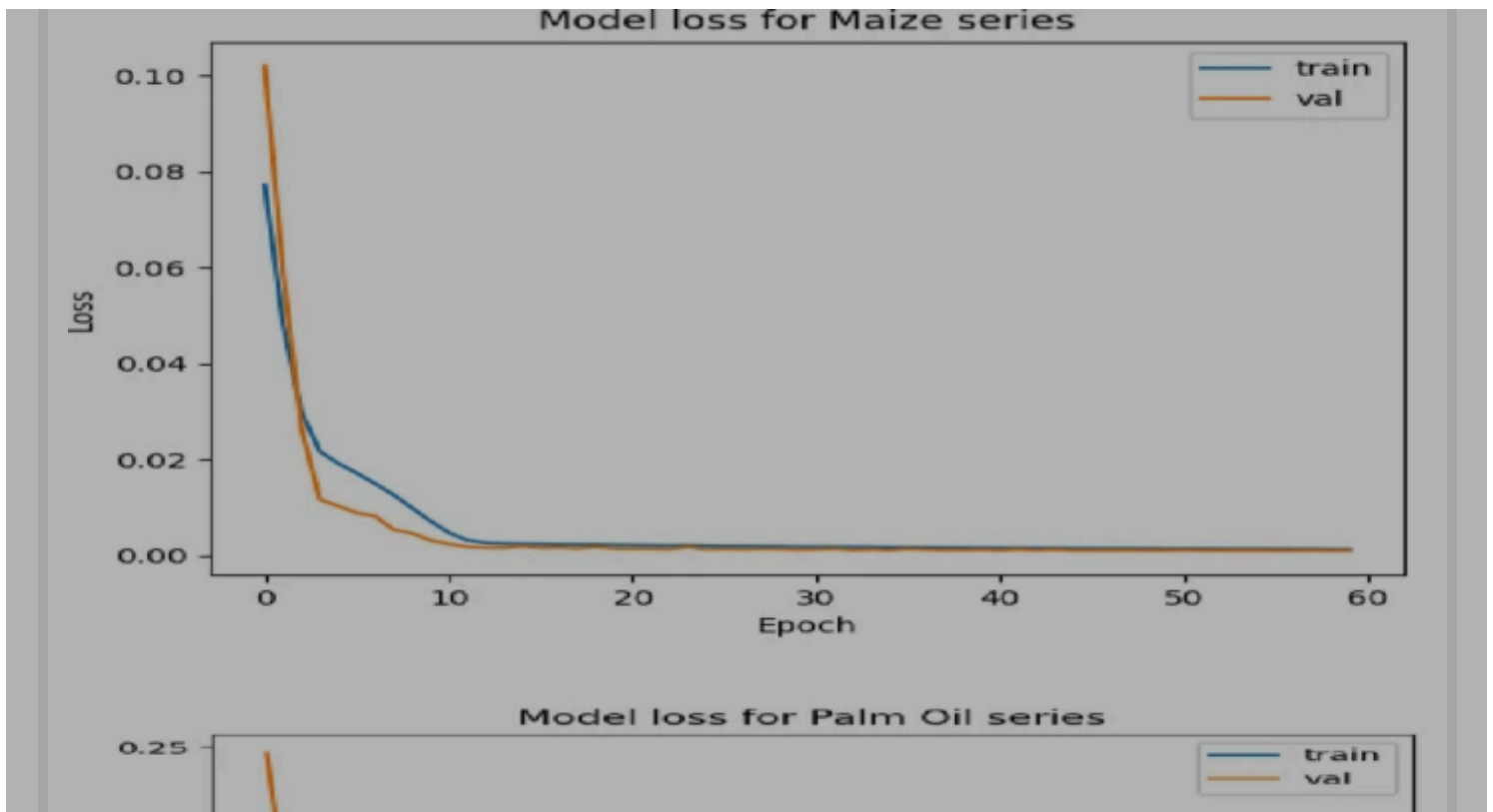
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Learning curves of the genetic algorithm optimized long short-term memory model for training and validation sets.

Fig. 6

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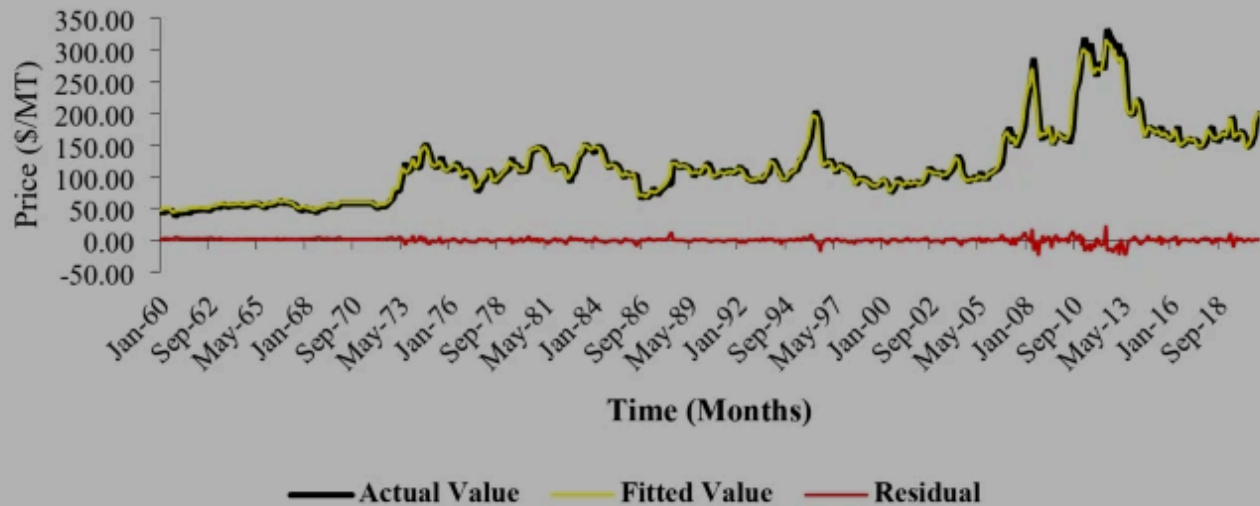
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Maize



Palm oil

1380.00

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The fitted value with observed series and residual of maize, palm oil, and soybean oil price for training set from genetic algorithm optimized long short-term memory model.

Table 9 Model hyperparameters utilized by the genetic algorithm optimized long short-term memory model for different price series.

- 1) The “decomposition-and-ensemble” hybrid models (EMD-LSTM, EEMD-LSTM, CEEMDAN-LSTM, and VMD-LSTM) consistently outperform the standalone LSTM by decomposing complex price series into simpler, more stable IMFs. The decomposition mitigates noise and isolates distinct patterns within the data, allowing the LSTM to model and predict each component more effectively. In

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demonstrate that hybrid models achieve lower RMSE and MAPE values, higher $\{D\}_{stat}$ scores, and significant DM statistics reflecting improved forecasting accuracy and better capture of both short-term fluctuations and long-term trends. The TOPSIS analysis ranks the standalone LSTM lowest, further reinforcing the advantage of decomposition-based approaches. The primary reason for decomposition improves forecasting accuracy is that agricultural price series are often characterized by sharp fluctuations, irregular patterns, and overlapping trends, making them highly volatile and complex. When a standalone LSTM model is applied directly to such data, it faces difficulty in distinguishing between meaningful patterns and random noise. This can result in inaccurate forecasts, as the model may either overfit to noise or miss critical underlying trends. Decomposition overcomes this challenge by disintegrating the original price series into simpler, more stable intrinsic mode functions (IMFs). Each IMF represents

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model, i.e. CEEMDAN-LSTM, then there is again a reduction in level measure MAPE (20.25%, 1%, 3.28%) and RMSE (8.22%, 1.89%, 5.38%) in comparison to EEMD-LSTM for maize, palm oil, and soybean oil, respectively. Finally, for the proposed VMD-LSTM, the reduction in MAPE (44%, 21.67%, 25.85%) and RMSE (56.93%, 21.83%, 27.00%) compared to CEEMDAN-LSTM. Table [10](#) also shows that the directional prediction statistics $\{D_{stat}\}$ observed maximum for maize through the CEEMDAN-LSTM model, palm oil through VMD-LSTM, and soybean oil through EEMD-LSTM and VMD-LSTM. Among hybrid models, TOPSIS ranks first to the proposed VMD-LSTM and fourth to EMD-LSTM. Table [11](#) shows that among hybrid models, VMD-LSTM performs better than all EMD-based LSTM models at the significance level of less than 1% for maize, same for palm oil except for EMD-LSTM is significant at 5%.

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Table 10 Forecasting performance of different models for maize, palm oil, and soybean oil price.

Table 11 Forecasting performance in terms of Diebold-Mariano test of different models for maize, palm oil, and soybean oil price for the one-year forecast horizon.

Fig. 7

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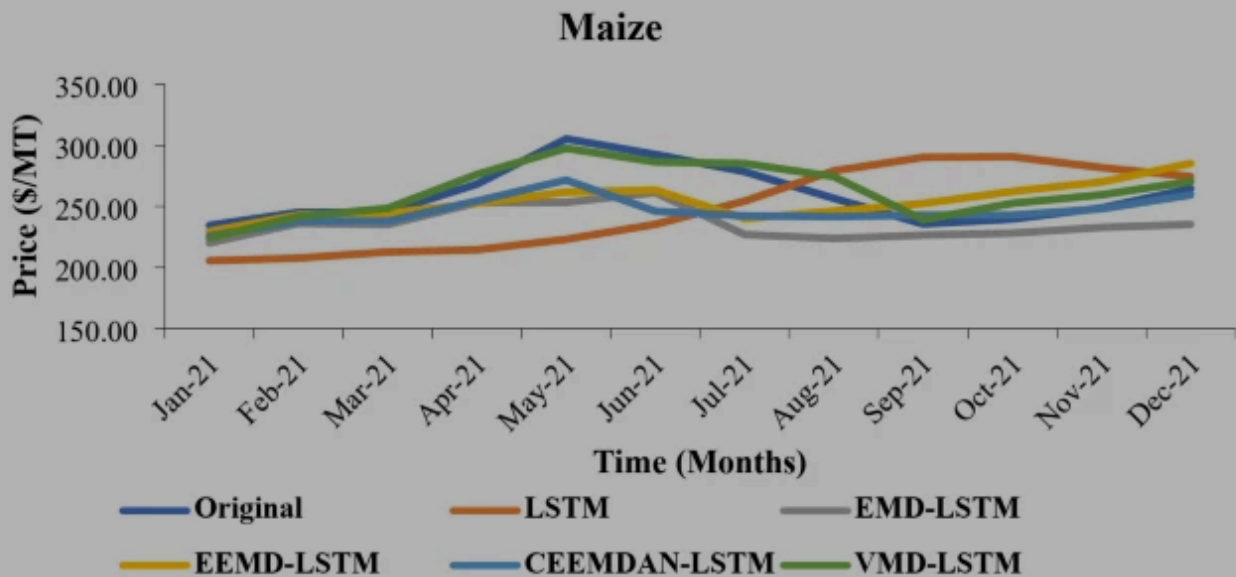
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Palm oil

1400.00

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Predicted results of different models.

Fig. 8



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Agriculture commodity prices play a critical role for all stakeholders involved in the agriculture industry, and price fluctuations can have far-reaching impacts on the national economy. To improve the forecasting accuracy of agriculture price, this paper proposed a novel GA-optimized VMD-LSTM forecasting model. The model utilizes the GA-optimized VMD technique, a powerful decomposition method that decomposes the original price series into multiple subseries. This decomposition reduces the volatility of the original price data, facilitating more effective analysis and forecasting. Subsequently, a GA-based LSTM network is utilized to create individual forecasting models for each independent mode obtained from the decomposition. These individual forecasts are then combined to produce an ensemble result. To assess the performance of the proposed model, experiments are conducted using monthly price data for maize, palm oil, and soybean oil from the international commodity market. The forecasting results of the proposed VMD-LSTM model are compared with LSTM, EMD-LSTM, FEEMD-LSTM,

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Contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Kapil Choudhary, Girish Kumar Jha, Ronit Jaiswal, and Rajeev Ranjan Kumar. The first draft of the manuscript was written by Kapil Choudhary. Girish Kumar Jha, Ronit Jaiswal, and Rajeev Ranjan Kumar commented on its improvement. Reviewing is done by Rajeev Ranjan Kumar and Girish Kumar Jha. All authors read and approved the final manuscript.

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